STAT-S 650 Time Series Analysis

Homework 1

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$$X = \begin{bmatrix} 2 & 1 & 8 \\ 3 & 1 & 7 \\ 5 & 4 & 5 \\ 3 & 5 & 9 \\ 6 & 8 & 2 \end{bmatrix}, \; Y = \begin{bmatrix} 3 & 2 \\ 4 & 2 \\ 5 & 5 \\ 6 & 4 \\ 5 & 7 \end{bmatrix}, \; J = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Practical

Compute the following steps and interpret each step in terms of a statistical concept

(Do some by hand to get the feel for it, check answers with Matlab or R, and then just do the rest in Matlab or R)

1. $M_X = (J' \cdot X)/5$
2. $M_Y = (J' \cdot Y)/5$
3. $D_X = X - J \cdot M_X$
4. $D_Y = Y - J \cdot M_Y$
5. $C_{XX} = (D'_X \cdot D_X)/4$
6. $C_{YY} = (D'_Y \cdot D_Y)/4$
7. $S_X = \sqrt{diag\,(C_{XX})}$
8. $S_Y = \sqrt{diag\,(C_{YY})}$
9. $R_{XX} = S_X^{-1} \cdot C_{XX} \cdot S_X^{-1}$ (note S_X must be a diagonal matrix)
10. $R_{YY} = S_Y^{-1} \cdot C_{YY} \cdot S_Y^{-1}$ (note S_Y must be a diagonal matrix)
11. $C_{XY} = (D'_X \cdot D_Y)/4$
12. $B = C_{XX}^{-1} \cdot C_{XY}$
13. $Y_p = D_X \cdot B$
14. $E = D_Y - Y_p$
15. $SSE = E' \cdot E$

In [3]:
```
%load_ext rpy2.ipython
#Loading R Cell Magic in a Jupyter Notebook
```

In [5]:
```
%%R

# Matrix X
X <- matrix(c(
```

```
  2, 1, 8,
  3, 1, 7,
  5, 4, 5,
  3, 5, 9,
  6, 8, 2
), nrow = 5, byrow = TRUE)


# Matrix Y
Y <- matrix(c(
    3,2,
    4,2,
    5,5,
    6,4,
    5,7
), nrow = 5, byrow = TRUE)

# Matrix J
J <- matrix(1, nrow = 5, ncol = 1)
```

In [7]:
```
%%R
# 1. Mx = (J' * X)/5
M_X <- (t(J) %*% X) / 5
print(M_X)
```

```
     [,1] [,2] [,3]
[1,]  3.8  3.8  6.2
```

In [8]:
```
%%R
# 2. My = (J' * Y)/5
M_y <- (t(J) %*% Y)/5
print(M_y)
```

```
     [,1] [,2]
[1,]  4.6    4
```

In [11]:
```
%%R
# 3. Dx = X-J * Mx
jmx <- (J %*% M_X)
dx <- (X-jmx)
print(dx)
```

```
     [,1] [,2] [,3]
[1,] -1.8 -2.8  1.8
[2,] -0.8 -2.8  0.8
[3,]  1.2  0.2 -1.2
[4,] -0.8  1.2  2.8
[5,]  2.2  4.2 -4.2
```

In [12]:
```
%%R
# 4. Dy = Y -J * My
jmy <- (J %*% M_y)
dy <- (Y-jmy)
print(dy)
```

```
     [,1] [,2]
[1,] -1.6   -2
[2,] -0.6   -2
[3,]  0.4    1
[4,]  1.4    0
[5,]  0.4    3
```

In [13]:
```
%%R
# 5. Cxx = (D'x * Dx)/4
cxx <- (t(dx) %*% dx)/4
print(cxx)
```

```
     [,1]  [,2]  [,3]
[1,]  2.70  3.95 -4.20
[2,]  3.95  8.70 -5.45
[3,] -4.20 -5.45  7.70
```

In [14]:
```
%%R
# 6. Cyy = (D'y * Dy)/4
cyy <-(t(dy) %*% dy)/4
print(cyy)
```

```
     [,1] [,2]
[1,]  1.3  1.5
[2,]  1.5  4.5
```

In [20]:
```
%%R
# 7. Sx = Root(diag(Cxx))
sx <- sqrt(diag(cxx))
print(sx)
```

```
[1] 1.643168 2.949576 2.774887
```

In [21]:
```
%%R
# 8. Sy = Root(diag(Cyy))
sy <- sqrt(diag(cyy))
print(sy)
```

```
[1] 1.140175 2.121320
```

In [23]:
```
%%R
# 9. Rxx = S^-1x * Cxx * S^-1x
sx_inv <- 1/diag(sx)
rxx <- sx_inv %*% cxx %*% sx_inv
print(rxx)
```

```
     [,1] [,2] [,3]
[1,]  NaN  NaN  NaN
[2,]  NaN  NaN  NaN
[3,]  NaN  NaN  NaN
```

In [24]:
```
%%R
# 10. Ryy = S^-1y * Cyy * S^-1y
sy_inv <- 1/diag(sy)
ryy <- sy_inv %*% cyy %*% sy_inv
print(ryy)
```

```
     [,1] [,2]
[1,]  Inf  Inf
[2,]  Inf  Inf
```

In [25]:
```
%%R
# 11. Cxy = (D'x * Dy)/4
cxy <- (t(dx) %*% dy)/4
print(cxy)
```

```
     [,1]  [,2]
[1,]  0.9  3.25
[2,]  2.4  6.00
[3,] -0.4 -4.75
```

In [26]:
```
%%R
# 12. B = C^-1xx * Cxy
cxx_inv <- 1/cxx
b <- cxx_inv %*% cxy
print(b)
```

```
           [,1]      [,2]
[1,]  1.0361664  3.853643
[2,]  0.5771047  2.384000
[3,] -0.7066007 -2.491610
```

In [28]:
```
%%R
# 13. Yp = Dx * B
yp <- dx %*% b
print(yp)
```

```
          [,1]       [,2]
[1,] -4.752874 -18.096655
[2,] -3.010107 -11.751402
[3,]  2.206741   8.091104
[4,] -2.114890  -7.198623
[5,]  7.671129  28.955576
```

In [29]:
```
%%R
# 14. E = Dy - Yp
e <- dy - yp
print(e)
```

```
          [,1]       [,2]
[1,]  3.152874  16.096655
[2,]  2.410107   9.751402
[3,] -1.806741  -7.091104
[4,]  3.514890   7.198623
[5,] -7.271129 -25.955576
```

In [31]:
```
%%R
# 15. SSE = E' * E
sse <- (t(e) %*% e)
print(sse)
```

```
         [,1]      [,2]
[1,]  84.2373  301.0931
[2,] 301.0931 1129.9880
```

In [35]:
```
%%R
# 16. Eigen values, Eigen Vectors and V * L * V' compared with Cxx
eigen_CXX <- eigen(cxx)

V <- eigen_CXX$vectors
L <- diag(eigen_CXX$values)

vlvt <- V %*% L %*% t(V)
```

In [36]:
```
%%R
cat("Eigenvalues (diagonal of L):\n")
print(diag(L))
```

```
Eigenvalues (diagonal of L):
[1] 16.1353707  2.7890851  0.1755442
```

In [37]:
```
%%R
cat("\nEigenvectors (columns of V):\n")
print(V)
```

```
Eigenvectors (columns of V):
           [,1]       [,2]       [,3]
[1,] -0.3937119 -0.1390581  0.9086549
[2,] -0.6702615  0.7199034 -0.1802461
[3,]  0.6290791  0.6800015  0.3766398
```

In [38]:
```
%%R
cat("V * L * V\n")
print(vlvt)
```

```
V * L * V
      [,1]  [,2]  [,3]
[1,]  2.70  3.95 -4.20
[2,]  3.95  8.70 -5.45
[3,] -4.20 -5.45  7.70
```

In [40]:
```
%%R
cat("Cxx from above\n")
print(cxx)
```

```
Cxx from above
      [,1]  [,2]  [,3]
[1,]  2.70  3.95 -4.20
[2,]  3.95  8.70 -5.45
[3,] -4.20 -5.45  7.70
```